Algebra 1 Name Answer Key ______________
Unit 1 Date______________________
Lesson 9 Practice Period______________________

1. Which of the following expressions can be combined?

A. $2\sqrt{80}-6\sqrt{20}$ $2\sqrt{16\cdot 5}-6\sqrt{4\cdot 5}$
B. $3\sqrt{6}+2\sqrt{12}$ $2\cdot 4\sqrt{5}-6\cdot 2\sqrt{5}$
C. $5\sqrt{18}-18\sqrt{5}$ $8\sqrt{5}-12\sqrt{5}$
D. $-4\sqrt{50}+5\sqrt{40}$ $-4\sqrt{5}$

Three students simplified the expression $2\sqrt{90}-7\sqrt{40}$ below. Their work is shown below.

Amy	**Linda**	**Jaxson**
$2\sqrt{90}-7\sqrt{40}$ $2\sqrt{(9\cdot 10)}-7\sqrt{(4\cdot 10)}$ $6\sqrt{10}-14\sqrt{10}$ $-8\sqrt{10}$	$2\sqrt{90}-7\sqrt{40}$ $-5\sqrt{50}$	$2\sqrt{90}-7\sqrt{40}$ $2\sqrt{(9\cdot 10)}-7\sqrt{(4\cdot 10)}$ $18\sqrt{10}-28\sqrt{10}$ $-10\sqrt{10}$

2. Which student simplified the expression correctly?
Amy is correct.

3. Write a note to one of the students who made an error explaining their mistake.
Answers will vary.
Linda, good try, but the expressions inside the radical cannot be subtracted.
You would first need to simplify both radicals and combine if they are like.

4. Write a note to the other student who made an error explaining their mistake.
Answers will vary.
Jaxson, you started this correctly by identifying perfect squares under the radical sign – but you forgot to take the principal square root when you brought them out.
So, you should have multiplied 2 by 3 (and not 9) and 7 by 2 (and not 4).

For question 5-10, find the sum or difference of the expression. If you cannot add or subtract the expression, explain why.

5. $\sqrt{18} - 4\sqrt{2}$

$\sqrt{9 \cdot 2} - 4\sqrt{2}$

$3\sqrt{2} - 4\sqrt{2}$

$\boxed{-\sqrt{2}}$

6. $\sqrt[3]{128} - \sqrt[3]{54}$

$\sqrt[3]{64 \cdot 2} - \sqrt[3]{27 \cdot 2}$

$4\sqrt[3]{2} - 3\sqrt[3]{2}$

$\boxed{\sqrt[3]{2}}$

7. $7\sqrt{47} + 8\sqrt{147}$

$7\sqrt{47}+56\sqrt{3}$

47 is prime and therefore must stay under the radical sign. $147 = 49 \cdot 3$, so $8\sqrt{147} = 8\sqrt{49 \cdot 3} = 8 \cdot 7\sqrt{3}$.

$\boxed{7\sqrt{47} \text{ and } 56\sqrt{3} \text{ are not like and cannot be combined (not like terms).}}$

8. $125^{\frac{1}{2}} + 320^{\frac{1}{3}}$

$\sqrt{125} + \sqrt[3]{320}$

$\sqrt{25 \cdot 5} + \sqrt[3]{64 \cdot 5}$

$5\sqrt{5} + 4\sqrt[3]{5}$

$\boxed{\text{These cannot be added since the indices are different (not like terms).}}$

9. $\sqrt{200} - 2\sqrt{48} + 4\sqrt{98}$

$\sqrt{100 \cdot 2} - 2\sqrt{16 \cdot 3} + 4\sqrt{49 \cdot 2}$

$10\sqrt{2} - 2 \cdot 4\sqrt{3} + 4 \cdot 7\sqrt{2}$

$10\sqrt{2} + 28\sqrt{2} - 8\sqrt{3}$

$\boxed{38\sqrt{2} - 8\sqrt{3}}$

10. $192^{\frac{1}{3}} + 2\left(81^{\frac{1}{3}}\right)$

$\sqrt[3]{192} + 2\sqrt[3]{81}$

$\sqrt[3]{64 \cdot 3} + 2\sqrt[3]{27 \cdot 3}$

$4\sqrt[3]{3} + 2 \cdot 3\sqrt[3]{3}$

$\boxed{10\sqrt[3]{3}}$

$4\sqrt[3]{3} + 6\sqrt[3]{3}$